Sample Question Paper

CBSE · Class 10 · Mathematics

— TOPPER-STYLE ANSWER BOOKLET · EVALUATED —

TOTAL: 8 / 8

SECTION A

Q1. The HCF of 12 and 18.

Find the prime factorisation of 12: $12 = 2 \times 2 \times 3 = 2^2 \times 3^1$ ✓ **+0.25**

Find the prime factorisation of 18: $18 = 2 \times 3 \times 3 = 2^1 \times 3^2$ ✓ **+0.25**

The HCF (Highest Common Factor) is the product of the common prime factors raised to the lowest power of their occurrence. HCF(12, 18) = $2^1 \times 3^1 = 6$ ✓ **+0.25**

∴ **(c) 6 Reason: The HCF of 12 and 18 is 6, as it is the largest number that divides both 12 and 18 exactly. ($12 = 6 \times 2$, $18 = 6 \times 3$).** ✓

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Examiner: Answer is perfectly correct with clear working and reasoning.

Q2. Find the roots of $x^2 - 5x + 6 = 0$.

Given quadratic equation: $x^2 - 5x + 6 = 0$. We will solve this by factorisation method. ✓

Split the middle term (-5x) such that the product of the coefficients of the split terms equals the product of the coefficient of x^2 (1) and the constant term (6). Here, -2 and -3 satisfy this condition ($-2 + -3 = -5$ and $-2 \times -3 = 6$). **+0.75**

$x^2 - 2x - 3x + 6 = 0$ ✓

Factorise by grouping: $x(x - 2) - 3(x - 2) = 0$ ✓ **+0.5**

$(x - 2)(x - 3) = 0$ ✓ **+0.25**

Set each factor to zero to find the roots: $x - 2 = 0 \Rightarrow x = 2$ $x - 3 = 0 \Rightarrow x = 3$ ✓ **+0.5**

∴ **The roots of the equation $x^2 - 5x + 6 = 0$ are 2 and 3.** ✓

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Examiner: Steps are clear and logical, leading to the correct roots. Full marks awarded.

SECTION B

Q3. State and prove the Basic Proportionality Theorem (draw the figure).

Statement of Basic Proportionality Theorem (Thales Theorem): If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then the other two sides are divided in the same ratio. ✓ +0.5

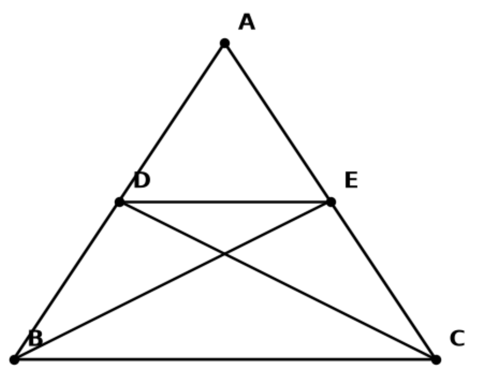
Figure: (See associated figure) ✓ +0.5

Given: In ΔABC, D is a point on AB and E is a point on AC such that DE $\parallel$ BC. **To Prove:** AD/DB = AE/EC **Construction:** Join BE and CD. Draw EM $\perp$ AB and DN $\perp$ AC. ✓ +0.5

Proof: 1. Consider the area of ΔADE and ΔDBE. Area(ΔADE) = ½ × base × height = ½ × AD × EM Area(ΔDBE) = ½ × base × height = ½ × DB × EM Ratio of their areas: Area(ΔADE) / Area(ΔDBE) = (½ × AD × EM) / (½ × DB × EM) = AD/DB ... (1) ✓ +0.75

2. Similarly, consider the area of ΔADE and ΔDEC. Area(ΔADE) = ½ × base × height = ½ × AE × DN Area(ΔDEC) = ½ × base × height = ½ × EC × DN Ratio of their areas: Area(ΔADE) / Area(ΔDEC) = (½ × AE × DN) / (½ × EC × DN) = AE/EC ... (2) ✓ +0.75

3. Triangles ΔDBE and ΔDEC are on the same base DE and between the same parallel lines DE and BC. Therefore, their areas are equal. Area(ΔDBE) = Area(ΔDEC) 4. From equations (1) and (2), using Area(ΔDBE) = Area(ΔDEC): Area(ΔADE) / Area(ΔDBE) = Area(ΔADE) / Area(ΔDEC) AD/DB = AE/EC Hence Proved. ✓ +0.5



∴ The Basic Proportionality Theorem is stated and proved, showing that if a line parallel to one side of a triangle intersects the other two sides, then it divides the two sides proportionally. ✓

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Examiner: Excellent presentation. The statement is precise, figure is clear, and proof is well-structured with all necessary steps and justifications. Full marks.

Q4. Define photosynthesis.

Definition of Photosynthesis: Photosynthesis is the metabolic process by which green plants, algae, and some bacteria convert light energy, usually from the sun, into chemical energy in the form of glucose (sugar). This process uses carbon dioxide and water as raw materials, with the help of a green pigment called chlorophyll, and releases oxygen as a byproduct. +1.5

Balanced Chemical Equation for Photosynthesis: 6CO■ + 6H■O + Light Energy → C■H■■O■ + 6O■ (Carbon Dioxide + Water + Light Energy → Glucose + Oxygen) +0.5

∴ **Photosynthesis is the process by which green plants and certain other organisms transform light energy into chemical energy, converting carbon dioxide and water into sugars and oxygen.**

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Examiner: A comprehensive definition including raw materials, products, energy source, and the chemical equation. Perfectly answered for full marks.

— Examiner's overall remark —

Excellent presentation. Step-wise working shown clearly — keep diagrams labelled and conclusions highlighted.